Passenger Preference Analysis in the Domestic Airline Industry

Submitted By

Devansh Dewan, CB.SC.U4CSE23017

Yoganand, CB.SC.U4CSE23253

Naman Yadav, CB.SC.U4CSE23338

Madhumithraa T S, CB.SC.U4CSE23429

Vrithika K M, CB.SC.U4CSE23457

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Under the Guidance of

Mr. Vedaj Padman

Assistant Professor (Sl.Gd.)

School of Computing, Coimbatore

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AMRITA SCHOOL OF COMPUTING

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE - 641 112

ABSTRACT

The rapid growth in India's domestic air travel industry has greatly increased the need for passenger expectations and behavior. An airline operates in a highly competitive market where customer satisfaction, pricing strategies, quality of service, and reliability are key factors in attracting and retaining customers. Understanding passenger expectations can help airlines improve service quality, operational efficiency, and the overall passenger flying experience.

The current project focuses on analyzing passenger expectations in India's domestic air travel industry using a data-driven approach. This study combines primary research in the form of a structured survey, along with supplementary information from publicly available airline sources. Approximately 2,100 passenger responses were collected from people with varying age groups, professions, and travel frequencies. The survey includes various attributes such as travel purpose, travel frequency, preferred flight class, booking behavior, last flown airline, and key decision drivers such as airline selection, ticket price, punctuality, airline brand, and in-flight amenities.

Before performing the analysis, various preprocessing steps have been followed. These preprocessing steps have included addressing missing values, normalizing the passenger responses, converting multi-label attributes into structured attributes, and addressing any imbalances in the categories. After preprocessing, various exploratory data analysis techniques have been performed to analyze passenger behavior and airline preferences.

The results show that various trends exist in passenger behavior. Ticket price, punctuality, and airline brand have a significant effect on passenger behavior. Travel purpose and occupation have a strong correlation with travel frequency and class. These results provide valuable insights that can help airlines improve passenger flying experiences in the domestic air travel industry.

In conclusion, the current project highlights the use of data analytics in analyzing passenger behavior and creating valuable insights that can help make informed decisions in the airline industry.

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1 Introduction

India's domestic air transport market is growing rapidly with the growth of cities, the economy, and air travel. With the growing competition, passenger demands shape the quality of air service. Air transport planning involves factors such as the cost of tickets, time performance, safety, quality of service, and reputation.

This project utilizes data to find out what passengers want from domestic air carriers. It examines the data, which consists of 2,099 responses to a survey from people with varied ages, jobs, and travel habits. It contains information about reasons for travel, frequency, class, booking methods, last-used airline, and what is important when choosing an airline.

Once the data were collected, cleaning and preparation of the data for analysis followed. Exploratory data analysis involved an examination of how passengers' characteristics relate to their travel preferences, thus providing an understanding of what affects their choice of an airline.

1.1 Scope and Limitations

The scope of this particular project is to analyze the passenger preference within the domestic air industry. To carry out this analysis, the study will be based on the data obtained from surveys. The analysis will be based on the factors that influence the choice of an airline, such as the price of the ticket, punctuality, service, safety, and the booking habits. Additionally, the analysis will be based on the influence of other factors such as the age group, occupation, and the frequency of travel. These factors will be analyzed to gain further insights on the subject.

The limitations are as follows: (1) a sample size of approximately 2,099 online survey results, which may not be representative of all domestic airline passengers; (2) based on personal preference rather than actual behavior; (3) results are not based on extensive prediction models.

2 Objectives

Key Project Goals:

- To determine the factors influencing passenger airline choices for domestic air travel
- To understand the relationship between passenger demographics, frequency, and airline choices..
- To understand passenger booking habits, purpose of travel, and class choices.

- To visualize passenger airline choices using data analysis techniques.
- To understand passenger airline choices to improve customer satisfaction.

3 Team Structure and Roles

Roll Number	Role/Name	Responsibilities
CB.SC.U4CSE23338	Scrum Master/Naman Yadav	Sprint planning, timeline coordination, milestone tracking, documentation, and final integration.
CB.SC.U4CSE23457	Data Engineer/Vrithika	Survey design, data collection, preprocessing, data cleaning, and dataset preparation.
CB.SC.U4CSE23253	Data Analyst/ Yoganand	Exploratory data analysis, visualization, trend identification, and insight generation.
CB.SC.U4CSE23017	Machine Learning Engineer/Devansh	Clustering, segmentation, and predictive modelling for passenger behaviour analysis.
CB.SC.U4CSE23429	Business Analyst / Madhumithraa	Converting analytical findings into business insights and recommendations.

Table 1: Team Roles and Responsibilities

4 Methodology

The study uses data analytics to assess the passenger's preference and sentiment with regard to air travel. Data analytics entails structured data from surveys and unstructured data from airline reviews. The study entails various techniques such as data preprocessing, exploratory data analysis, prediction modeling, customer segmentation, and sentiment analysis.

4.1 Data Preprocessing

The survey dataset has been cleaned by handling missing values, standardizing the data, and encoding the categorical features. The features like influencing factors and reward preferences are multi-select features, which were binarized using

the 'MultiLabelBinarizer' class, and the numerical values were scaled using the 'StandardScaler' class.

For the review data set, preprocessing steps included lowercasing, removal of punctuation and special characters, removal of stopwords using NLTK, and lemmatizing.

4.2 Descriptive Analytics

EDA involved analyzing passenger demographics, travel habits, preferred airlines, and factors affecting the selection of the airline. We used Pandas, Matplotlib, and Seaborn libraries to create the visualizations.

4.3 Predictive Modelling

The data has been split in the ratio 80:20 to train the models and test the models. Logistic Regression and Random Forest classifiers were used to classify airline preference and passenger loyalty. Accuracy and confusion matrix were used to evaluate the models.

4.4 Customer Segmentation

K-Means clustering method grouped the passengers based on similar behavior. The optimal number of clusters was determined by using the Elbow Method.

4.5 Text Mining and Sentiment Analysis

The reviews were converted to numerical features using TF-IDF Vectorization. Sentiment analysis identified the reviews as positive, negative, or neutral.

4.6 Tools and Technologies

The analysis was performed by using Python with libraries such as Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn and NLTK

5 Data Sources

For analyzing the passenger preferences and sentiment, the study relies on two types of data. Both structured and unstructured data have been used in the study. While structured data helps in collecting measurable travel patterns and passenger preferences, unstructured data helps in collecting passenger experiences.

The two sets of data used in the research are as follows:

The research utilizes survey data from airline passengers, which includes their travel patterns, airline preferences, and the factors that affect their choice of airline services.

The study uses text data in the form of customer reviews related to airline services, covering all aspects of the services in which passengers are involved.

By analyzing two sets of data, the research is able to look at the airline service from different perspectives. The survey data allows the research to look at the passenger's point of view, while the textual data allows the research to look at the customer's point of view.

5.1 Survey Dataset

The survey collected passenger preference on airline travel behavior from a variety of people, such as students, professionals, and business travelers. The aim of the survey was to identify passenger airline choice behavior. The survey received 2100 online responses, where people shared their experiences.

The survey database has various fields that indicate the behavior of passengers in traveling and the preferences of passengers in choosing an airline. The survey helps us understand how passengers choose an airline.

The dataset contains a number of variables that describe passenger airline choice behavior. The key attributes included in the survey dataset are listed below:

- Gender
- Age group
- Occupation
- Travel frequency within the past 12 months
- Purpose of travel (business, leisure, education, or family visit)
- Preferred airline
- Preferred booking platform
- Preferred travel class
- Flight scheduling preferences
- Factors influencing airline choice such as:
  - Ticket price
  - Punctuality

  - Safety rating
  - Brand reputation
  - In-flight service quality

The dataset includes structured numerical and categorical data, which can be used to analyze passenger preferences and behaviors, for instance, how they rate the services of the airlines and what factors affect their choices of airlines.

5.2 Airline Review Dataset

In addition to the survey, the current research will also utilize the existing database of customer reviews of various airlines. The reviews include the written feedback of various passengers based on their personal experiences with various airlines. The feedback of the customers is a significant way of gaining insights into the perception of the customers about various aspects of the service. The reviews are considered an effective method of extracting the sentiments of the customers, which may not be possible through other means of survey.

The source of the reviews is publicly available airline rating websites. In these reviews, the customers willingly express their experiences and opinions based on the service they have experienced from the airlines, as they have traveled with the airlines. This represents the experiences of the customers in terms of the good and bad, as they have experienced the service from the airlines.

The reviews of passengers include their experience of different flights. They include their experience of the crew's service, the comfort of the seats, the punctuality of the flights, and the handling of their baggage, and so on. They also include their opinions on the booking and checking in, and other facilities like food and entertainment.

Typical review topics include:

- Flight punctuality and delays
- Cabin crew service and staff behavior
- Seat comfort and legroom
- Baggage handling and airport services
- In-flight services and amenities
- Boarding and check-in experience
- Overall passenger satisfaction

5.3 Primary Dataset

The basic dataset used for the text mining part includes the airline passenger reviews, along with the airline information and the rating score. This dataset is used as the primary source to analyze the customer sentiment and identify the patterns from the passenger reviews.

In the original dataset, there were several attributes related to airline review. However, in this case, only the features that are directly related to text mining and sentiment analysis are kept. This helped in keeping the dataset simple and only the relevant information is kept.

The main dataset revolves around a certain set of attributes considered as the primary variables :

- **Review_Text**:The comments of the travelers about their experiences with the airline.
- **Airline**:The airline that is linked to the passenger review allows for the examination of customer sentiment for that particular airline.
- **Rating**: A numerical value that represents how content a passenger is with the airline's service.

5.4 Attempted Supplementary Dataset

In addition to the main data set, the researchers examined additional opinions from airline customers on various online airline forums and travel discussion sites. These sites are where people discuss and share tips, praise, or complain about airline service with other consumers. The intention was to increase the overall range of customer feedback and gain a better understanding of the common issues and desires of consumers. However, most of these sites have restrictions on automated data collection and web scraping in an effort to preserve user-generated content. Therefore, it was not possible to gather a large amount of reviews from these sites. The additional set of comments helped in understanding the overall theme in the feedback of the flyers.

5.5 Merged Dataset

Finally, the text mining dataset was constructed using the most important features retained from the initial review dataset, along with other analytical features added to the dataset. This process aimed to transform the raw and unstructured data into a clean and well-structured dataset, which could then be readily utilized for sentiment analysis and topic identification. During the preprocessing step, the

data points containing missing or incorrect values were eliminated, and the original text of the reviews was normalized using a set of normalization techniques. Only the features required for the analysis were retained, and the unnecessary features were filtered out to minimize the size of the data and maintain the focus.

The final dataset includes the following attributes:

- **Review_Text**: Cleaned textual feedback written by passengers describing their airline travel experiences.
- **Airline**: The airline associated with each passenger review, enabling airline-wise analysis of customer feedback.
- **Rating**: A numerical score representing the passenger's level of satisfaction with the airline service.
- **Sentiment (Derived)**: A sentiment label generated from the review text using sentiment analysis techniques.

The sentiment score was based on text reviews, and the reviews were classified as positive, negative, or neutral to evaluate the general sentiment of the customers and compare the trends of different airlines. This data acts as a source for the text mining process, and the general sentiment and recurring problems of the airlines can be analyzed.

6 Exploratory Data Analysis (EDA)

EDA was conducted to understand the data's structure, identify any inconsistencies, and prepare the data for analysis. Preprocessing and transformation were conducted to ensure clean data for machine learning.

6.1 Removing Duplicate Rows

Duplicate rows were removed to prevent bias, and the response would map to a unique passenger by removing the duplicates.

```
print("Duplicate rows:", df_clean.duplicated().sum())
print(df_clean.isnull().sum())
```

Figure 1: Identifying and deleting duplicate record

6.2 Renaming Column Name Aliases

Some column names had inconsistent naming patterns or aliases. The column names are standardized to improve readability. This also makes it easier to manipulate the data.

```python
df = df.rename(columns={
    "Gender": "Gender",
    "Age": "Age",
    "Occupation": "Occupation",
    "Purpose of Travel": "Purpose_of_Travel",
    "Travel Frequency (Past 12 months)": "Travel_Frequency",
    "Preferred Travel Class": "Travel_Class",
    "Direct Flight or Layover Preference": "Flight_Preference",
    "How do you usually Book your flights ?": "Booking_Mode",
    "What influences your choice of airline the most ?": "Influencing_Factors",
    "Which airline have you flown in last 12 months": "Airline_Last_Flown",
    "How sensitive are you to price changes": "Price_Sensitivity",
    "Are you a part of any Airline Loyalty program?": "Loyalty_Program",
    "What type of reward do you value most?": "Reward_Preference",
    "What type of schedule do you prefer": "Schedule_Preference",
    "Which In-flight services do you prioritize the most": "Inflight_Priority"
})
```

Figure 2: Standardization of column names

6.3 Converting Records with Multiple Values to List Format

Some questions allowed multiple values (for influencing factors or service options). These questions were converted to a structured list format to facilitate proper processing and feature transformation..

```python
for col in multi_select_cols:
    df[col] = (
        df[col]
        .fillna("")
        .apply(lambda x: [i.strip() for i in x.split(";") if i])
    )
```

Figure 3: Conversion of multi-value responses into structured list format

6.4 Encoding Loyalty Program

The loyalty program attribute was transformed into a machine learning-compatible numeric format, where categorical response types such as membership were represented as binary/categorized numbers.

```python
df["Loyalty_Program"] = df["Loyalty_Program"].map({
    "yes":1,
    "no":0
})

df["Loyalty_Program"] = df["Loyalty_Program"].fillna(0)
```

(a) Encoding of loyalty program variable

```python
multi_cols = [
    "Influencing_Factors",
    "Reward_Preference",
    "Inflight_Priority"
]

for col in multi_cols:

    df[col] = df[col].apply(parse_list)

    mlb = MultiLabelBinarizer()

    encoded = pd.DataFrame(
        mlb.fit_transform(df[col]),
        columns=[col + "_" + c for c in mlb.classes_],
        index=df.index
    )

    df = pd.concat([df, encoded], axis=1)

    df = df.drop(columns=[col])
```

(b) Multi-label binarization of categorical attributes

Figure 4: Encoding and multi-label binarization techniques

6.5 Feature Engineering – Removing Unused Columns

After preprocessing and encoding, redundant columns were eliminated as they did not contribute to the modelling process. Feature engineering helped in ensuring that only relevant columns were present in the final dataset.

```python
drop_cols = [
    "Purpose_List",
    "Airline_Last_Flown",
    "airline_count",
    "indigo",
    "spicejet",
    "vistara",
    "air_india",
    "akasa",
    "others"
]

df = df.drop(columns=drop_cols)
```

Figure 5: Final feature selection after removing unused columns

7 Data Insights

In this section, the major descriptive findings obtained from the data exploration phase are presented, which include demographic information, market share, and other price sensitivity trends. Furthermore, a thorough text mining analysis on the reviews provided by the passengers will be conducted to identify the major

phrases responsible for both positive and negative sentiments for each airline.

7.1 Demographic Insights

The airline can offer more tailored services and marketing strategies based on their understanding of the passengers' demographics.

7.1.1 Gender Distribution

The gender distribution, based on the percentage of male and female passengers, helps the airline tailor their loyalty programs.

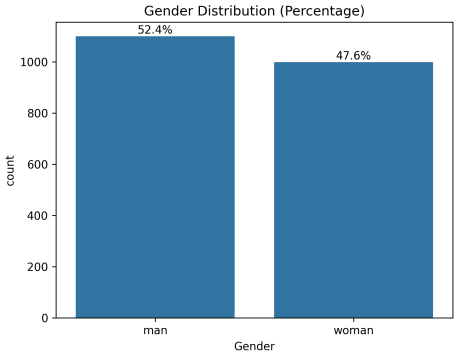

Figure 6: Gender Distribution of Customers

7.1.2 Age Group Distribution

The majority of the passengers are comprised of the working professional group, as they are between the ages of 25 and 34 and 35 and 44. The implication is that there are opportunities for offering premium services, as they are economically active.

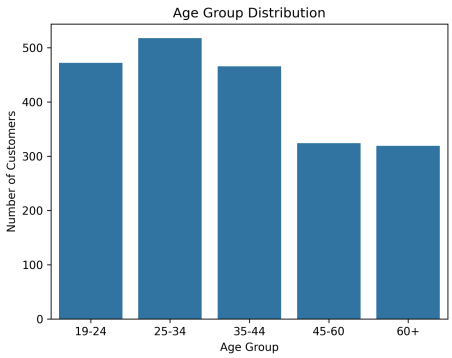

Figure 7: Age Group Distribution of Customers

7.1.3 Occupation Distribution

The majority of passengers are working professionals and business owners, highlighting the significance of business travel demand. There is also a sizable student population, which suggests that there is a need for affordable fares.

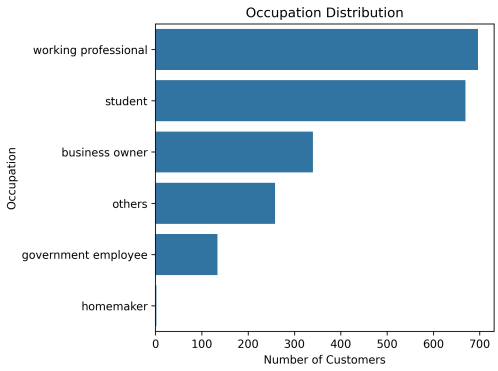

Figure 8: Occupation Distribution

7.2 Travel Preferences and Market Share

A clear picture of the competitive landscape can be obtained by comprehending market distribution and travel frequency.

7.2.1 Travel Frequency

By understanding the travel frequency of the customers, it is easier to understand the target audience for the milestones of the frequent flyer program and seasonal campaigns.

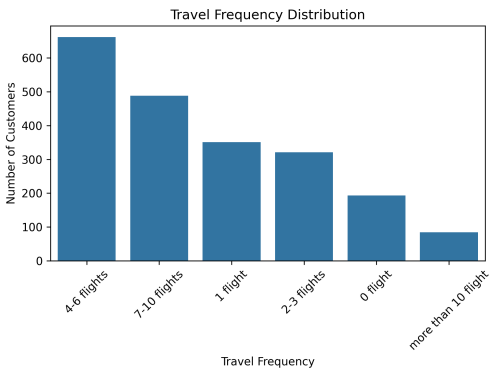

Figure 9: Travel Frequency of Customers

7.2.2 Airline Market Share

The market share of Indigo is the highest among all the airlines, followed by Vistara and Spicejet. This represents the general trend for the Indian aviation

industry.

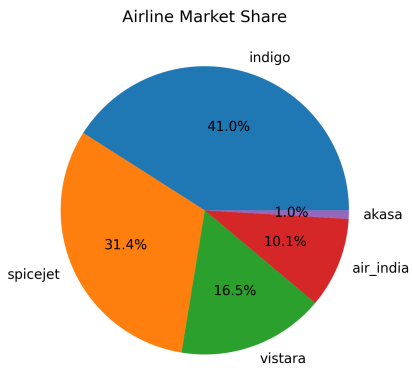

Figure 10: Airline Market Share

7.2.3 Age-wise Airline Preference

The visualization represents the preferred choice of airlines for people belonging to different age groups. This shows the preference for legacy or low-cost carriers based on age groups.

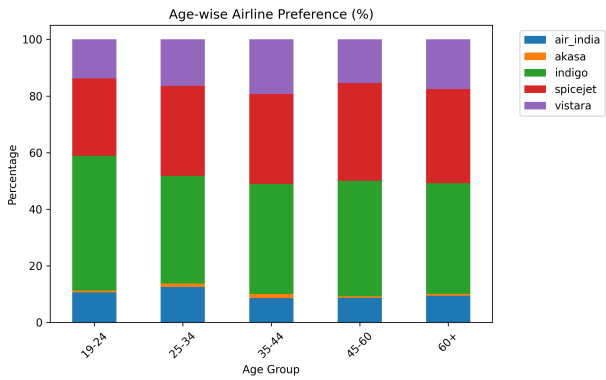

Figure 11: Age-wise Airline Preference

7.3 Price Sensitivity Analysis

The price sensitivity for passengers varies across different airlines. Full-service airlines tend to carry a higher number of passengers with low price sensitivity when compared to low-cost carriers.

7.3.1 Overall Price Sensitivity

The chart below represents the price sensitivity for the overall passenger base.

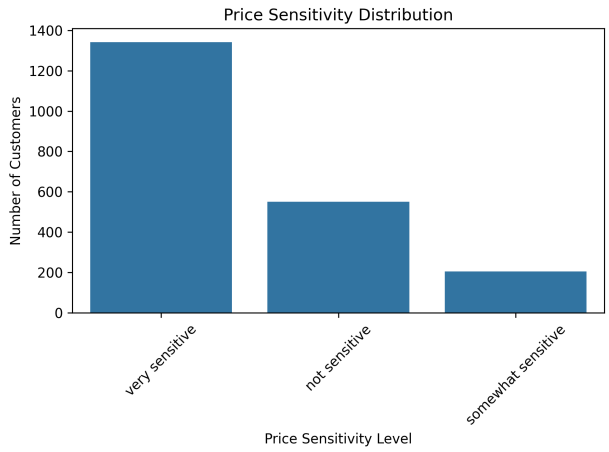

Figure 12: Price Sensitivity Overview

7.3.2 Airline vs Price Sensitivity

The figure shows the comparison of price sensitivity for customers of various major airlines.

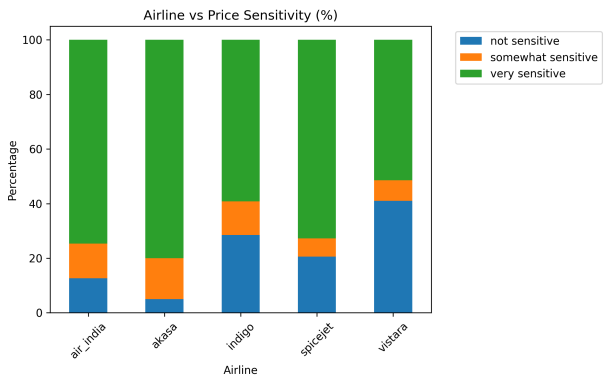

Figure 13: Price Sensitivity segmented by Airline

Top Customer Phrases (Text Mining Insights)

To identify the main factors affecting customer emotions among all airlines, a frequency analysis of certain key phrases from customer reviews was conducted. The above graphs represent the positive and negative key phrases for all major airlines.

Indigo Insights

The passengers of Indigo are highly appreciative of its punctuality and affordability; however, they are not too happy with their connecting flights and customer service.

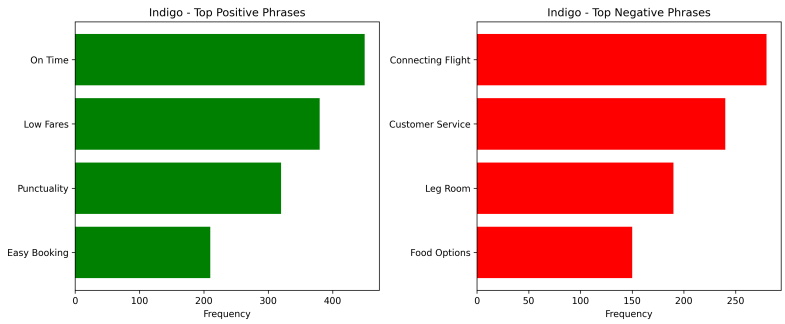

Figure 14: Top Customer Phrases for Indigo

SpiceJet Insights

While 'Low Fares' and 'Friendly Crew' are the major factors for SpiceJet's high-rated reviews, flight service disruptions such as 'Flight Delay' and 'Cancellation' are major negative feedback points for SpiceJet.

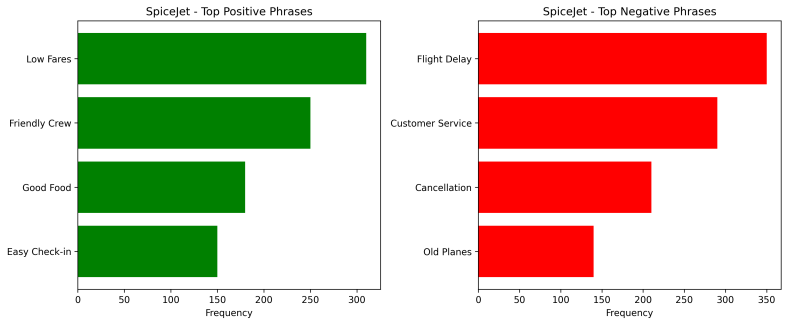

Figure 15: Top Customer Phrases for SpiceJet

Vistara Insights

Vistara's high-rated reviews are based on its premium service, with 'Premium Cabin' and 'Good Food' being the major positive feedback points. However, negative feedback points for Vistara include 'High Prices.'

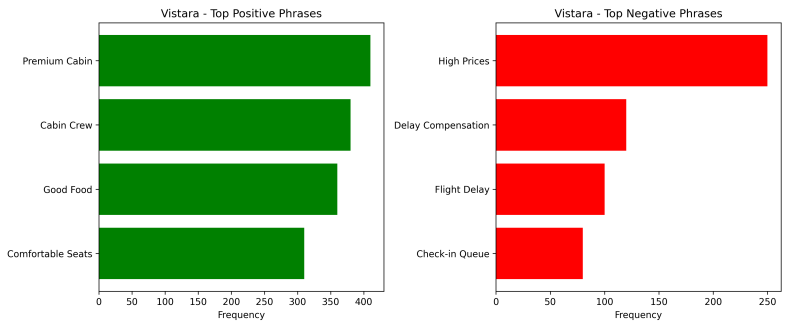

Figure 16: Top Customer Phrases for Vistara

Air India Insights

Baggage allowance and route options have been the two biggest strengths Air India has to offer. On the flip side, the biggest weaknesses Air India faces are with regards to the airline's 'Old Planes' and the quality of their 'Customer Service'.

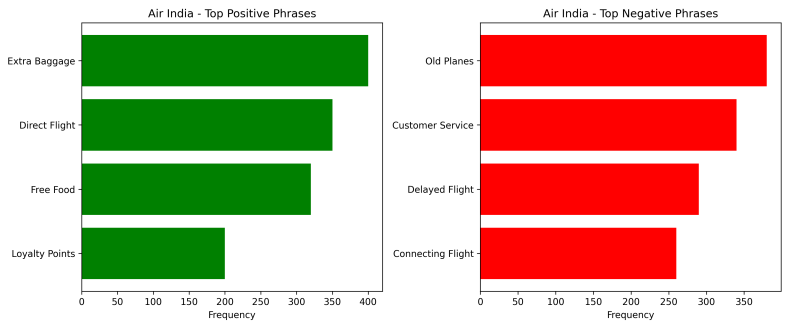

Figure 17: Top Customer Phrases for Air India

Akasa Insights

Akasa Air, as a relatively new airline, has managed to win the trust of customers with regards to their 'New Planes'. However, the biggest weakness Akasa faces is with regards to their 'Limited Routes'.

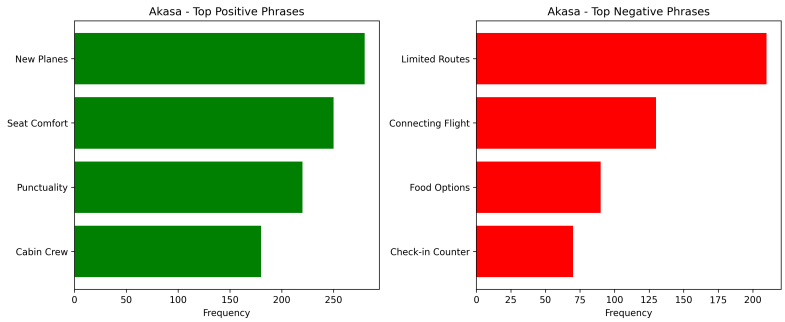

Figure 18: Top Customer Phrases for Akasa

Strategic Recommendations Based on Text Mining

- **Addressing Pain Points:** Airlines must lay more emphasis on the improvement of the *Customer Service* process during the 'Flight Delays' and 'Connecting Flights' scenarios.
- **Capitalizing on Strengths:** Promotional activities must incorporate terms like *Punctuality* and *Good Food*, as these attributes strongly resonate with customer satisfaction and positively influence airline preference.

8 Customer Segmentation

Customer segmentation is carried out in order to identify different segments of airline passengers based on their traveling habits, service requirements, and loyalty levels. The passengers are divided into segments based on their attributes, and then the airline can understand their needs and develop marketing strategies accordingly.

8.1 Passenger Segmentation using Clustering

Cluster analysis is one of the methods that can be employed to classify passengers based on certain factors that are relevant and significant in nature. Cluster analysis helps in identifying the common needs that passengers share, such as the needs of frequent flyers, price-conscious passengers, and service-conscious passengers.

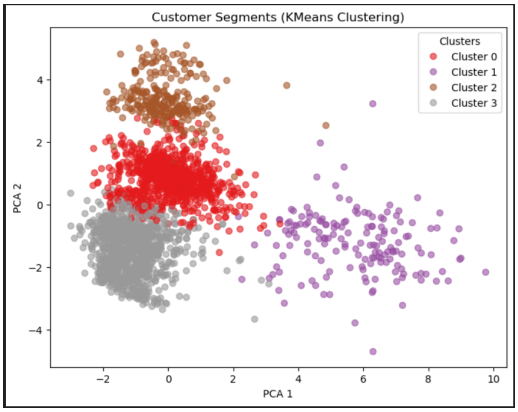

Figure 19: Customer segmentation results using clustering

9 Predictive Modelling

Predictive modeling helped identify the patterns that influence passenger loyalty, airline choice, and sentiment. Various machine learning algorithms were used to develop classification models and test predictive performance.

9.1 Loyalty Program

Logistic Regression

Logistic Regression was used as a baseline to predict whether a customer will enroll or continue to use an airline loyalty program.

```
Logistic Regression Accuracy: 0.9452380952380952

Confusion Matrix
[[278   8]
 [ 15 119]]

Classification Report
              precision    recall  f1-score   support

         0.0       0.95      0.97      0.96       286
         1.0       0.94      0.89      0.91       134

    accuracy                           0.95       420
   macro avg       0.94      0.93      0.94       420
weighted avg       0.95      0.95      0.94       420
```

(a) Logistic Regression model

```
Random Forest Accuracy: 0.9476190476190476

Confusion Matrix
[[277   9]
 [ 13 121]]

Classification Report
              precision    recall  f1-score   support

         0.0       0.96      0.97      0.96       286
         1.0       0.93      0.90      0.92       134

    accuracy                           0.95       420
   macro avg       0.94      0.94      0.94       420
weighted avg       0.95      0.95      0.95       420
```

(b) Random Forest model

Figure 20: Models used for loyalty program prediction

Top Factors Influencing Loyalty

The Feature Importance analysis revealed the key factors influencing passenger participation in the airline's loyalty program.

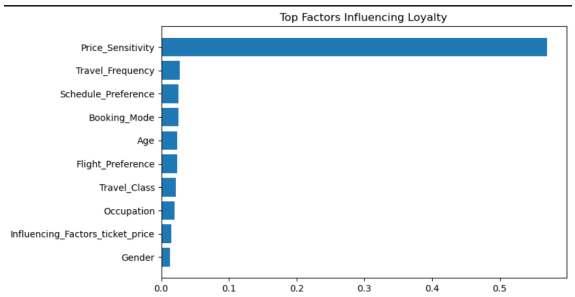

Figure 21: Top factors influencing airline loyalty

9.2 Airline Flight Preference

Decision Tree

The Decision Tree model revealed the underlying rules for passenger airline loyalty.

```
Decision Tree Accuracy: 0.3142857142857143
[[ 7  0 14 18 10]
 [ 1  0  1  2  1]
 [20  1 62 54 32]
 [15  4 54 44 20]
 [ 5  0 19 17 19]]
              precision    recall  f1-score   support

           0       0.15      0.14      0.14        49
           1       0.00      0.00      0.00         5
           2       0.41      0.37      0.39       169
           3       0.33      0.32      0.32       137
           4       0.23      0.32      0.27        60

    accuracy                           0.31       420
   macro avg       0.22      0.23      0.22       420
weighted avg       0.32      0.31      0.32       420
```

Figure 22: Decision Tree model for airline preference prediction

Random Forest

The Random Forest model improved the model's capabilities by using a combination of multiple decision trees.

```
Random Forest Accuracy: 0.3761904761904762
[[  0   0  32  17   0]
 [  0   0   4   1   0]
 [  0   0 118  48   3]
 [  0   0  95  37   5]
 [  1   0  42  14   3]]
              precision    recall  f1-score   support

           0       0.00      0.00      0.00        49
           1       0.00      0.00      0.00         5
           2       0.41      0.70      0.51       169
           3       0.32      0.27      0.29       137
           4       0.27      0.05      0.08        60

    accuracy                           0.38       420
   macro avg       0.20      0.20      0.18       420
weighted avg       0.31      0.38      0.31       420
```

Figure 23: Random Forest model for airline preference prediction

Top Factors Influencing Airline Choice

Feature importance analysis was conducted for major airlines such as Air India, Akasa Air, IndiGo, SpiceJet, and Vistara.

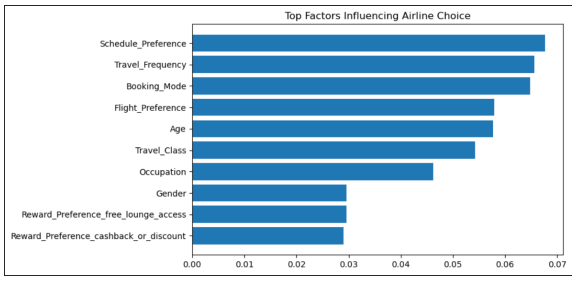

Figure 24: Top factors influencing airline choice

9.3 Text Mining

Sentiment Analysis

Text mining techniques were used to analyze the airline reviews, where NLP techniques were applied.

```
Airline distribution:
 Airline
Indigo       10000
Air India     8822
Akasa         3287
SpiceJet       876
Vistara        186
Name: count, dtype: int64

Sentiment distribution:
 Sentiment
Positive    12423
Negative    10257
Neutral       491
Name: count, dtype: int64
```

Figure 25: Sentiment analysis results from airline reviews

Sentiment Prediction

Textual features of the passenger reviews were used for training the machine learning models for sentiment prediction.

```
Accuracy: 0.853258523953388

Classification Report:

              precision    recall  f1-score   support

    Negative       0.81      0.88      0.84      2051
    Positive       0.90      0.83      0.86      2583

    accuracy                           0.85      4634
   macro avg       0.85      0.86      0.85      4634
weighted avg       0.86      0.85      0.85      4634
```

Figure 26: Sentiment prediction model results

10 Business Insights

10.1 Occupational Trends

The customer base of the airline under consideration comprises mainly Working Professionals and Students. Together, they make the largest segment of the customer base. About 700 customers are Working Professionals, and around 670 customers are students. This shows a division of the market between "Time-Critical" customers and "Budget-Critical" customers.

10.2 Age & Price Sensitivity

The age group between 25 and 34 years comprises the largest customer segment of the airline under consideration. This shows the peak age of professional mobility. The age groups 19–24 years and 35–44 years follow the largest customer segment. Price sensitivity is high among all the customers. More than 60% of all

the age groups are "Very Sensitive" to the price of the tickets. The age group 60+ years shows the highest price sensitivity among all the customers. About 67.4% of the customers falling under this age group are "Very Sensitive" to the price of the tickets. The age group 45–60 years shows the highest number of customers "Not Sensitive" to the price of the tickets. About 29.0% of the customers falling under this age group are "Not Sensitive".

10.3 Customer Segmentation (KMeans Clustering)

The clustering analysis shows the existence of four distinct types of travelers based on their behavioural and demographic characteristics. These are:

1. **Cluster 0 (Red) - Budget Multi-Purpose Travelers:** This is the largest cluster of travelers. The customers have varied travel purposes. However, they are moderately price-sensitive.

2. **Cluster 1 (Purple) - Frequent Business Travelers:** This cluster of customers shows distinct characteristics. The customers are highly frequency-oriented. They have specific schedule requirements.

3. **Cluster 2 (Brown) - Service Travelers:** This group of customers is highly focused on the quality of the service provided in the flight.They are highly satisfied with the quality of the service provided in the flight.

4. **Cluster 3 (Grey) - Price-Sensitive Leisure Travelers:** This is the most price-sensitive group of customers. They are mainly focused on the lowest airfare

10.4 Predictive Drivers of Airline Preference & Loyalty

The predictive model shows the fundamental factors driving the decision-making of the customers. These are:

10.4.1 The Loyalty/Price Paradox

The predictive model shows that the Price Sensitivity factor dominates all other factors. This factor is the only driving force behind customer loyalty. This factor alone holds a significant weightage of more than 0.5. This shows the low brand loyalty of the customers. The brand loyalty of the customers is highly dependent on the price factor.

10.4.2 Predicting Airline Choice

The predictive model shows the fundamental factors driving the decision-making of the customers. These are:

1. **Predicting Airline Loyalty:** The initial decision to choose an airline is highly dependent on the Schedule Preference factor. This is followed by Travel Frequency. The customers are highly focused on the booking mode.

2. **Carrier-Specific Influencers:**

   - Akasa Air: This airline shows a high degree of influence from Ticket Price (25.6%).
   - Vistara: This shows a high degree of influence from Safety Rating (16.8%).
   - Air India & SpiceJet: These two airlines benefit the most from Loyalty Programs & Punctuality.

10.5 Customer Sentiment Trends

The sentiment analysis of over 23,000 customer reviews indicates that customer service is a dynamic area of focus. Out of all customer reviews, 54% are positive sentiment, whereas 45% are negative sentiment, indicating that service quality is a major determinant of brand preference.

10.6 Key Drivers of Customer Satisfaction (Text Mining)

The text mining analysis on over 23,000 customer reviews indicates that satisfaction is built on a foundation of both operational efficiency and interpersonal warmth.

- **Interpersonal Excellence:** Friendliness of the staff is the key influencer of positive sentiment, which indicates that interpersonal interaction is the most memorable part of the experience. When the crew exceeds the minimum expectations to deliver care, the impact is enormous.

- **Physical Foundations:** Seat comfort and cleanliness were identified as the "hygiene factors" of air travel.High-frequency positive words related to cabin upkeep and anatomical support indicate these as the bare minimum for a 5-star passenger experience.

- **Service Efficiency:** Smooth and fast processes, such as check-in, baggage handling, and in-flight meal service, function as immediate stress-relievers and correlate very well with high satisfaction scores.

10.7 Major Causes of Customer Dissatisfaction (Text Mining)

Operational instability and failures in professional conduct are the leading predictors of negative brand sentiment.

- **Operational Instability:** Delays and cancellations are identified as the most damaging. The predictive model identified that these are not just immediate causes of anger but long-term causes of a lack of trust, which overshadows other causes.
- **Professional Conduct Failure:** Staff rudeness is identified as a singular most identified emotional trigger for customer dissatisfaction.
- **Coefficient Analysis:** * The words ”rude” and ”unprofessional” are identified as having the highest coefficient for customer dissatisfaction.This is a clear area where there is a critical failure in terms of customer satisfaction, as it appears that a customer is more likely to rate an experience as poor if they do not see a level of empathy or professional etiquette in a stressful situation

11 Business Relevance & Strategic Recommendations

11.1 Multi-Tiered Market Segmentation & Targeted Growth

The four behavioural clusters of Budget Multi-Purpose, Frequent Business, Service-Focused, and Price-Sensitive Leisure offer a marketing roadmap for high-impact marketing strategies:

- **Targeting the “Gender Growth” Gap:** With males dominating 57% of passengers, this is a massive opportunity that needs to be filled by women. This could be done by offering “Safety & Comfort” packages for leisure travel, targeting women who prefer flying in Premium Economy (parity exists between classes).
- **Niche Capture for Business Owners:** Since Business Owners have a 38% chance of choosing Business Class regardless of price, this is an opportunity for offering “Corporate Mobility” memberships, targeting this niche group.

11.2 Dynamic Pricing & Revenue Optimization

The extremely price-sensitive nature of the market (60%+):

- **Low Price Anchoring:** The airline has to compete on price for the "Price-Sensitive Leisure" segment and the "Budget Seniors" (60+). The airline can use a "Low Fare Alert" system and offer an "Unbundled Economy" option to retain the bulk of this segment.

- **Premium Upselling for the 'Sweet Spot':** The 45–60 age group is the least price-sensitive segment (29% not price sensitive). The airline has to target this segment for premium upselling for Premium Economy and Business Class. The airline can use the benefits of "Executive Comfort" and "Hassle-Free" for these passengers.

- **Value-Based Scheduling:** As Schedule Preference is the key driver for airline selection, prime time flights should be sold at a premium. Frequent business travelers (Cluster 1) are time-sensitive to their schedules.

11.3 Reimagining Loyalty in a Price-Dominant Market

The data has clearly indicated a "Loyalty Paradox" wherein price is the most important loyalty driver (> 0.5 weight). Hence, the traditional loyalty program needs a relook:

- **Immediate Financial Gratification:** Instead of rewarding customers for point accumulation over a period of time, the program must introduce Instant Cashback or Companion Discounts since the market has clearly moved from economic incentives to experiential benefits.

- **Platform-Based Retention:** Since Booking Mode is a key predictor for customer loyalty, the airline must introduce "App-Only" loyalty tiers to reduce reliance on third-party booking engines.

- Incentivize customers for booking on the airline's own mobile app for loyalty benefits.

11.4 Operational Excellence & Brand Positioning

Text mining and predictive analysis show that operational problems are more important than marketing messages:

1. **Zero-Tolerance for operational "Rudeness":** Words like "rude" and "unprofessional" are the top factors affecting customer feelings. Therefore, staff need training in emotional intelligence to stop the brand from getting damaged.

2. **Competitive Differentiation:**

    - Akasa Air should focus on the "Budget Multi-Purpose" segment because price is a main reason customers stay loyal.
    - Vistara should use its high "Safety Rating" (16.8%)to attract customers who care about safety.
    - Air India and SpiceJet should focus on punctuality since their loyalty programs already have a strong impact on customer loyalty, and reliability is key for customer habits.

11.5 Strategic Resource Allocation

- **High ROI Service Areas:** The data from text mining shows that Seat Comfort, Cleanliness, and Staff Friendlyness are the top three service areas that matter most for customer satisfaction. Companies should focus their investments on improving these areas rather than on brand promotions, since these factors are closely linked to positive customer feedback(54% baseline).
- **Volatility Mitigation:** Since Leisure is a type of travel that happens during certain seasons, it's a good idea to invest in data related to Education and Family segments (36%) to help fill flights through student discounts and family group bookings.

12 Conclusion

The purpose of the study was to find what passengers prefer and how they feel about the airlines by using the survey results and reviews. The results of the study helped to have a better idea of how people travel, what factors influence their choices of airlines, and how they feel about the airlines' services.

By looking at the results, we could find trends about the people who travel, how often they travel, and their favorite airlines. We used techniques such as Logistic Regression and Random Forest to find the factors that influence passenger loyalties and their choices of airlines. We also used K-Means to cluster passengers according to their travel habits and expectations.

We read reviews for an airline to gain more information on what customers go through. The results included common problems and positive feedback, such as service quality, delays, and satisfaction.

The study revealed that factors such as price, on-time performance, and reputation are considered when customers select an airline. These findings will allow

an airline to gain more insights on what customers need and improve their services to make customers happier.

In the future, more data and advanced techniques such as improved natural language processing will be used to gain further insights on customer behavior.

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